Answers to Exercises

1. $2+2i = 8^{1/2}e^{i(\pi/4)} \implies (2+2i)^6 = \left(8^{1/2}e^{i(\pi/4)}\right)^6 = 8^{6/2}e^{i(6\pi/4)} = 8^3e^{i(3\pi/2)}$

2. $2+2i = 8^{1/2}e^{i(\pi/4)} \implies (2+2i)^{1/3} = \left(8^{1/2}e^{i(\pi/4)}\right)^{1/3} = 8^{1/6}e^{i(\pi/12+2\pi k/3)} \quad k = 0,1,2$

$$k=0: \quad 8^{1/6}e^{i(\pi/12)}$$
$$k=1: \quad 8^{1/6}e^{i(9\pi/12)}$$
$$k=2: \quad 8^{1/6}e^{i(17\pi/12)}$$

3. $4(1-\sqrt{3}i) = 8e^{i(5\pi/3)}$ *or* $8e^{i(-\pi/3)} \implies 4\left((1-\sqrt{3}i)\right)^7 = 2^2\left(2e^{i(-\pi/3)}\right)^7 = 2^9e^{i(-7\pi/3)}$

4. $4(1-\sqrt{3}i) = 8e^{i(5\pi/3)}$ *or* $8e^{i(-\pi/3)} \implies 4\left((1-\sqrt{3}i)\right)^{1/4} = 2^2\left(2e^{i(-\pi/3)}\right)^{1/4} = 2^{9/4}e^{i(-\pi/12+2\pi k/4)}$
$k = 0,1,2,3$

$$k=0: \quad 2^{9/4}e^{i(-\pi/12)}$$
$$k=1: \quad 2^{9/4}e^{i(5\pi/12)}$$
$$k=2: \quad 2^{9/4}e^{i(11\pi/12)}$$
$$k=3: \quad 2^{9/4}e^{i(17\pi/12)}$$

5. $-2+2i = 8^{1/2}e^{i(3\pi/4)}$ and $3i = 3e^{i(\pi/2)}$

$$\implies \left(\frac{-2+2i}{3i}\right)^{1/2} = \left(\frac{8^{1/2}e^{i(3\pi/4)}}{3e^{i(\pi/2)}}\right)^{1/2} = \left(\frac{8^{1/2}}{3}e^{i(\pi/4)}\right)^{1/2} = \frac{8^{1/4}}{3^{1/2}}e^{i(\pi/8+2\pi k/2)} \quad k = 0,1$$

$$k=0: \quad \frac{8^{1/4}}{3^{1/2}}e^{i(\pi/8)}$$
$$k=1: \quad \frac{8^{1/4}}{3^{1/2}}e^{i(9\pi/8)}$$

6. $z^{1/6} - 2 - 2i = 0 \implies z = (2+2i)^6 = 8^3e^{i(3\pi/2)} \quad (\mathit{from}\ 1.)$

7. $z^3 + 5i = 2+7i \implies z = (2+2i)^{1/3} = 8^{1/6}e^{i(\pi/12+2\pi k/3)} \quad k = 0,1,2 \quad (\mathit{from}\ 2.)$

8. $3iz^2 + 2 = 2i \implies z = \left(\dfrac{-2+2i}{3i}\right)^{1/2} = \dfrac{8^{1/4}}{3^{1/2}}e^{i(\pi/8+2\pi k/2)} \quad k = 0,1 \quad (\mathit{from}\ 5.)$